

SPENCER HOYUN KIM

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EDUCATION

Carnegie Mellon University | Pittsburgh, PA

Bachelor of Science in Mechanical Engineering & Biomedical Engineering | May 2027
Overall GPA: 3.32/4.0

Northview High School | Duluth, GA

High School Diploma | GPA 3.8/4.0 | May 2023

RESEARCH EXPERIENCE

Cook Cardiopulmonary Engineering Lab Undergraduate Student Researcher | CMU

May 2025 ~ Present

- Participating in a Rabbit PK/PD study aiming to study the effects of FXII900 on clotting time and the number of plasmas and platelets that exist after 30 minutes, 1 hour, 2 hours, 4 hours, 6 hours, 8 hours, and 24 hours, assisting in the coagulation testing and activated clotting time measurements.
- Participating in a 30-day sheep study on long-term ambulatory respiratory support for patients with chronic lung disease to test if the Pulmonary Assist System (PAS), when used with oral anticoagulation and biocompatible surface coatings, safely and effectively provide 30-day ambulatory respiratory support in an ovine model. Assisting with sheep surgery, blood sampling, sample testing, data analysis, and additional support that is needed during the 30-day study period.
- Participating in a 15-day sheep study on long-term ambulatory respiratory support for patients with chronic lung disease to test if the Pulmonary Assist System (PAS), when used with oral anticoagulation and biocompatible surface coatings, safely and effectively provide 15-day ambulatory respiratory support in an ovine model. Assisting with sheep surgery, blood sampling, sample testing, data analysis, and additional support that is needed during the 15-day study period.
- Participating in rabbit surgery testing the benefits of FXII900 on anticoagulation, conducting tests for CBC, ELISA, Coagulation (PT and aPTT), as well as charting the condition of the animal during the 2-hour surgery and 4-hour study period.

PUBLICATIONS & PATENTS

- Shin S, Liu D, Potchernikov A, Scala H, Bennett J, Nakamura R, Haggerty M, Kim S, Jiang S, Cook KE. Pharmacokinetic and Pharmacodynamic Profile Evaluation of Polycarboxybetaine-Conjugated FXIIa Inhibitor in Rabbits. *34th Congress of the International Society on Thrombosis and Haemostasis (ISTH)*. July 2026. Poster Presentation.
- Kumpfbeck, A. R., Bennett, J., Kelley, J. F., Bulard, B., Bapatla, S., Woo, Y., Petrovic, M., Kane, S., Said, B., Akhavanmalayeri, A., Kim, S., Shin, S., Dorken-Gallastegi, A., Strong, K., Scala, H., Demarest, C. T., Skoog, D. J., Bacchetta, M., Cook, K. E., & Ukita, R. "In vivo testing of a novel, low profile, and compact centrifugal pump in veno-venous and veno-arterial configurations of mechanical circulatory support." *AATS MCS Abstract Submission: VUMC-CMU-Inspired Therapeutics Pump Studies*.
- South Korea - CANE Patent (Patent Number: 10-2675388), June 2024

ACADEMIC & TECHNICAL PROJECTS

Assistive Page-Turning Device for Limited Hand Mobility (24-370 Mechanical Design Final Project) | CMU

Oct 2025 ~ Dec 2025

- Led user-driven redesign of a page-turning assistive device for readers with cerebral palsy
- Engineered a bidirectional, single-page isolation mechanism using a rubber wheel to separate the top sheet and a servo-actuated sweep arm to complete the flip; implemented large, low-force button with optional hands-free control
- Performed hand calculations and FEA of the assembly and subassembly to assess and de-risk performance and durability
- Delivered a manufacturable BOM and cost model (prototype built for \$115.80 under a \$200 budget; estimated unit COGS to be \$66.12 and target price of \$119.99 for small-batch validation)

Custom Wilderness Chair Design National Park Expedition Project | CMU

Feb 2025 ~ May 2025

- Designed a durable, wildlife-safe outdoor chair for climbers in Yosemite, emphasizing portability, weather resistance, and terrain stability
- Conducted user research to develop a cost-effective, multipurpose solution with quick setup and integrated gear storage for backcountry use

Bike Crank Arm Redesign for E-Bike Safety Project | CMU

Feb 2025 - Mar 2025

- Engineered a lightweight crank arm to protect riders by failing predictably at a 40 N pedal load, ensuring a factor of safety (FOS) near 1.0 at critical threshold
- Applied mechanical analysis to limit deflection under half-load to <7 mm (10% of span), ensuring rigidity during normal operation
- Optimized geometry and material use to minimize mass while maintaining structural integrity and ensuring failure occurs safely >1 cm from key interfaces
- Collaborated on safety-critical design to preserve product reliability and brand reputation for a startup e-bike company

Cane for the Visually impaired Project

May 2021 - June 2024

- Invented new cane design that can detect possible obstacles while a visually impaired person is walking and alert them by vibration.
- Designed cane to be more effective than a traditional cane while also avoiding damage to surrounding objects or people. Patented in Korea in 2024.

Concentrated Solar Power Plant Project | CMU

Mar 2025 - April 2025

- Simulated a Brayton cycle power plant in MATLAB, modifying turbine and compressor models to reflect 90% isentropic efficiency and evaluating system performance
- Analyzed effects of solar intensity, salt temperature, and pressure ratio on power output and efficiency; designed thermal storage to extend operation during low irradiance

Wind Turbine Blade Design Project | CMU

Feb 2025 - Mar 2025

- Optimized blade geometry using QBlade, MATLAB, and Solidworks to maximize turbine power output under design constraints
- Analyzed aerodynamic performance and delivered a technical report with simulation data and final design rationale

LEADERSHIP & STUDENT ORGANIZATIONS

American Society for Artificial Internal Organs – CMU Chapter (Member/Founding Member) | CMU

April 2025 ~ Present

- Contribute to the CMU student chapter of ASAIO, supporting interdisciplinary engagement in artificial organs, extracorporeal circulation, cardiovascular devices, and biomedical engineering.
- Participate in chapter activities connecting students with research, clinical translation, and professional development opportunities in cardiopulmonary and artificial organ technologies.
- Engage with peers and faculty to promote awareness of artificial internal organs, medical device innovation, and translational bioengineering.

Central Church of Pittsburgh Korean Ministry (President) | CMU

Aug 2025 ~ Present

- Lead a CMU Korean ministry community by organizing weekly gatherings, fellowship events, and student outreach.
- Coordinate communication between student members and Central Church of Pittsburgh leadership to support programming and engagement.
- Plan community-building activities that promote belonging, mentorship, and spiritual growth among students.

Carnegie Mellon Racing Club (Formula SAE) | CMU

Aug 2024 ~ May 2026

- Collaborated with a multi-disciplinary team to design and manufacture a fully electric and gasoline-powered formula-style race car for Formula SAE competitions.
- Designed the carbon monocoque and the chassis using Solidworks; performed FEA analysis using ANSYS Discovery to ensure structural integrity under race loads.
- Machined precision components using manual lathe, hand mill, band saw, and other machines, and utilized 3D printing for rapid prototyping of custom parts.

- Assisted in vehicle testing and validation, troubleshooting mechanical failures to improve reliability and performance for Formula SAE Competition.

WORK EXPERIENCE

Engineering Intern | XVision Technology, Seoul, South Korea

May - July 2024

- Inspired by my younger brother who is blind, started a project to create word files, which can be read aloud, on various subjects for easy access to those who are visually impaired. My word files also contain sound clips so that they can hear about the subjects, such as the wind on Mars.

Computing Services Department, Technology Enhanced Facilities Operator Manager | CMU

February 2024 ~ Present

- Monitor the school's computer labs as well as the printers and other computing services the school provides, assisting students in need of help with using the machines or devices located around the school related to the Computing Services Department.

TEACHING EXPERIENCE

Teaching Assistant | PSI Academy, Seoul, South Korea

May - August 2023

- Taught the basics of biology and robotics to help students who are participating in the 2023 iGEM and FIRST Global competitions.
- Taught the basics of synthetic biology, the experience that I have had in the field, and about what the team's topic would be and the necessary information related to the topic that the students would need to know.
- Introduced the process in robot building including the chassis, drivetrain, and other essential parts to the robot.

SKILLS

Software: MATLAB, Solidworks, AutoCAD Fusion360, Arduino, ANSYS Discovery, ANSYS Fluent, Microsoft Offices, Ultimaker Cura, PreForm, Microsoft Word, Microsoft PowerPoint, Microsoft Excel, Visual Studio Code, Python, Surgical Charting

Machines: Laser Cutter, 3D Printer, Lathe, Milling Machine, CNC Machine, Grinder, Band Saw, Hand Mill, Sand Blaster, Radiometer ABL 800 FLEX, Radiometer ABL 90 FLEX PLUS, Sysmex CA-600 Coagulation Analyzer Machine, ABAXIS Vetscan HM5

Languages: Fluent in Korean, Fluent in English